Project Name - Recruitment System

Problem 1 - Project Selection & Team Formation

1.1 Project Selection

The selected project is a Recruitment Management System designed to help organizations manage their internal hiring process. The system allows HR teams to manage job openings, store candidate information, track recruitment stages, schedule interviews, and generate recruitment reports.

The project was selected because:

• Recruitment processes in many organizations are still handled using spreadsheets and emails.

• HR teams need a centralized system to track candidate progress.

• The project can be realistically implemented within one academic semester.

• It demonstrates practical application of software engineering concepts.

1.2 Team roles

Role	**Name**
Product Manager	Teja Sreya
Lead Developer	Aishwarya Yalameli
QA Lead	Chintamreddy Vedapallavi
Documentation Lead	Thirisha N

2. Team Agreement (Working Norms)

Pennywise Team Working Agreement

Meeting Schedule:

- **Weekly Meetings:** Every Saturday, 5:00 PM - 6:00 PM (College Library,/Group Study Room/Virtual Meeting)
- **Emergency Meetings:** As needed, minimum 1 day notice via WhatsApp group
- **Duration:** Maximum 1 hour per meeting (we all have other subjects too)

Communication:

- **Primary:** WhatsApp group "SEAD Project"
- **Document Sharing:** Google Drive shared folder
- **Code Sharing:** GitHub repository
- **Response Time:** Within 24 hours for non-urgent, within 3 hours for urgent

Work Distribution Principles:

1. Everyone contributes equally
2. Tasks assigned based on expertise
3. Honest communication if workload gets too much

Decision Making:

- Minor decisions: Discuss in WhatsApp, majority votes wins
- Major decisions: Discuss in meetings, must reach consensus
- Tie-breaker: Project Manager has final call after hearing everyone

Quality Standards:

- Review each other's work before finalizing
- If something's not good enough, we redo it together

Conflict Resolution:

1. Talk it out directly first
2. If unresolved, involve the whole team
3. If still stuck, approach lab instructor
4. NO ghosting team members or passive aggression

What We Promise Each Other:

- Show up to meetings or inform in advance
- Complete assigned tasks on time or ask for help early
- Respect each other's ideas even if we disagree
- Have fun while learning - we're in this together!

Signatures: [All 4 members sign/date]

Problem 3 - Problem Statement Development

3.1 Problem Definition

Many organizations manage recruitment processes manually through spreadsheets, emails, and paper records. This often leads to scattered candidate information, difficulty tracking application stages, delayed interview scheduling, and inefficient communication between HR recruiters and hiring managers.

3.2 Affected Users

• HR Recruiters

• Hiring Managers

• HR Administrators

• Organization Management

3.3 Proposed Solution

The Recruitment Management System will provide a centralized platform where HR teams can manage job openings, store candidate details, track recruitment stages, schedule interviews, and generate hiring reports. The system improves efficiency, reduces manual errors, and enables better coordination between HR and hiring managers.

Problem 4 - Stakeholder Analysis

4.1 Primary Stakeholders

• HR Recruiters – Manage candidate applications and recruitment process.

• Hiring Managers – Review candidates and provide interview feedback.

4.2 Secondary Stakeholders

• HR Department Management

• Company Executives

4.3 Tertiary Stakeholders

• Development Team

• Project Instructor

Problem 5 - Stakeholder Interests and Influence

Stakeholder	Primary Interest	Influence
HR Recruiters	Efficient candidate management	High
Hiring Managers	Easy candidate evaluation	High
System Administrator	System stability and maintenance	Medium
Project Guide	Academic learning outcomes	High

Problem 6 - Stakeholder Register

ID	Stakeholder	Interest	Influence
STK-01	HR Recruiters	High	High
STK-02	Hiring Managers	High	High
STK-03	System Administrator	Medium	Medium
STK-04	Project Guide	High	High

Problem 7 & 8- Scope Definition – In Scope & Out of Scope

What's IN SCOPE (We WILL build)	**What's OUT OF SCOPE (We WON'T build)**
The system will include recruiter registration and login.	Not including a candidate-side mobile application.
Recruiters will be able to post and manage job openings.	AI-based resume screening.
Shortlist candidates, schedule interviews, update application status.	Video interview integration & advanced analytics dashboards.
Generate basic recruitment reports.	Multi-language support.
View and manage candidate applications.	Integration with external job portals.

Problem 9 - Scope Boundaries Document

• Internal recruitment system only

• Used by HR staff and hiring managers

• Candidate information manually entered by recruiters

Problem 10 -

10.1 Purpose

This document describes the functional and non-functional requirements of the **Recruitment Management System**. The purpose of this system is to provide a centralized platform for managing internal recruitment activities within an organization. It will guide the design, development, and testing of the system. The document defines system features, user roles, and operational requirements to ensure efficient recruitment workflow management.

10.2 Scope

The Recruitment Management System is designed to support the internal hiring process of an organization. The system will allow HR recruiters to create and manage job openings, store candidate information, track recruitment stages, schedule interviews, record feedback, and generate recruitment reports.

The system will centralize candidate data and improve coordination between HR recruiters and hiring managers. It will reduce manual work, minimize errors, and enhance recruitment efficiency by providing structured tracking of candidates across different hiring stages.

The system is intended for internal organizational use and will not include job seeker login functionality.

10.3 Definitions

Term	Meaning
User	A person who uses the system (HR Recruiter, Hiring Manager, Admin)
Admin	System administrator responsible for managing user roles and access
HR Recruiter	User responsible for managing job postings and candidate screening
Hiring Manager	Department head who reviews candidates and provides interview feedback
Dashboard	Main control screen displaying recruitment overview and statistics
Candidate	Individual applying for a job position
Job Opening	Available position created by HR for recruitment
Recruitment Pipeline	The stages a candidate goes through (Applied, Screening, Interview, Selected, Rejected)
Interview Feedback	Evaluation provided by interviewer after candidate interview
Report	System-generated summary of recruitment data

Problem 11

11.1 Product Perspective

The Recruitment Management System is designed to replace the current manual recruitment process, which typically involves managing candidate data through emails, spreadsheets, and paper records. These traditional methods often lead to data duplication, miscommunication, delayed interview scheduling, and difficulty in tracking candidate progress.

The proposed system will provide a centralized digital platform that connects HR recruiters, hiring managers, and administrators. It will allow users to manage job openings, store and update candidate information, track recruitment stages, schedule interviews, and generate reports. The system improves coordination between departments and ensures structured and efficient recruitment workflow management.

11.2 User Characteristics

User Type	Computer Skills	Special Needs
HR Recruiter	Basic to Good	Simple interface for managing candidates and updating status quickly
Hiring Manager	Basic to Good	Easy access to candidate details and ability to provide feedback

Admin	Good	Detailed controls for user management, data security, and system configuration

11.3 Constraints (Limitations)

- The system will run on college lab computers.
- The system will use only free and open-source tools.
- No budget is allocated for paid services or cloud hosting.
- The system will function primarily in an internal/local environment.
- Development time is limited to the academic semester duration.

Problem 12

3.1 Functional Requirements

FR-01: User Registration

Description: New recruiters/admin can create accounts
Priority: High
Acceptance Criteria:

1. User can enter name, email, and password
2. System validates email format
3. Password must be at least 8 characters
4. User account is created successfully

FR-02: User Login

Description: Registered users can log into the system
Priority: High
Acceptance Criteria:

1. User can log in using email and password
2. System verifies credentials correctly
3. Invalid login shows error message
4. User is redirected to dashboard after login

FR-03: Resume Upload

Description: Recruiters can upload candidate resumes
Priority: High
Acceptance Criteria:

1. System accepts PDF and DOC/DOCX files
2. File size is validated before upload
3. Uploaded resumes are stored successfully
4. Upload confirmation is shown

FR-04: Resume Parsing

Description: System extracts structured data from resumes
Priority: High
Acceptance Criteria:

1. System extracts name, skills, education, and experience
2. Extracted data is displayed in structured format
3. System handles missing fields without crashing

FR-05: Job Description Management

Description: Recruiters can create and manage job descriptions
Priority: High
Acceptance Criteria:

1. Recruiter can add job title, skills, and experience required
2. Job descriptions are saved in the system
3. Recruiter can edit or delete job descriptions

FR-06: Resume-Job Matching

Description: System matches resumes with job descriptions using ML model
Priority: High
Acceptance Criteria:

1. System generates a matching score for each resume
2. Same input produces consistent results
3. Matching is based on skills and experience

FR-07: Candidate Ranking

Description: System ranks candidates based on matching score
Priority: High
Acceptance Criteria:

1. Candidates are sorted in descending order of score
2. Top candidates are highlighted
3. Ranking updates automatically after new data

FR-08: Search and Filter Candidates

Description: Recruiters can search and filter candidates
Priority: Medium
Acceptance Criteria:

1. Recruiter can search candidates by skills
2. Filters (experience, education) refine results
3. Results update dynamically

FR-09: Results Display

Description: System displays candidate details and results
Priority: High
Acceptance Criteria:

1. Candidate name, skills, and score are displayed
2. Selection status (selected/rejected) is shown
3. Data is presented in a clear format

FR-10: Report Generation

Description: System generates reports for recruitment analysis
Priority: Medium
Acceptance Criteria:

1. System generates list of top candidates
2. Reports include selection statistics
3. Reports can be downloaded or viewed

Problem 13

3.2 Non-Functional Requirement

NFR-01: Performance - Response Time

Description: System should respond quickly to user actions
Requirement: Pages should load within 3 seconds for 95% of requests
Test: Measure using browser developer tools (Network tab)

NFR-02: Performance - Resume Processing

Description: Resume parsing should be efficient
Requirement: Each resume should be processed within 5 seconds
Test: Upload resumes and measure processing time

NFR-03: Security - Authentication

Description: Ensure secure access to the system
Requirement: Only authenticated users can access the system
Test: Try accessing system without login and verify restriction

NFR-04: Security - Data Protection

Description: Protect sensitive user and candidate data
Requirement: Passwords must be stored in encrypted format
Test: Verify database does not store plain text passwords

NFR-05: Accuracy - ML Model Performance

Description: System should provide accurate matching results
Requirement: Model should achieve at least 80% accuracy
Test: Evaluate using test dataset and calculate performance metrics

Problem 14 - Use case diagram

Problem 15 - Glossary

Term	Definition
Candidate	Individual applying for a job position
Job Opening	Available role created by HR
Recruitment Pipeline	Stages through which candidates progress
Interview Feedback	Evaluation given after candidate interview

16. Project Charter

The project charter defines objectives, stakeholders, roles, scope, and constraints for Recurtment syste.

17. Stakeholder Register

A formal stakeholder register is maintained as a separate artifact.

18. Complete SRS Document

This document constitutes the complete Software Requirements Specification for Pennywise.

19. Requirements Sign-Off

Name	Role	Signature	Date
Project Guide	Approver		

EXERCISE 2……..

Exercise 3: Agile Project Setup

38. Agile Board Setup

An Agile project board is established using Trello to effectively manage and track the development process of the Recruitment Management System. The board is structured to reflect the workflow of tasks from initiation to completion.

The following columns are defined:

- **Backlog** – Contains all identified user stories and tasks yet to be started
- **To Do** – Tasks selected for the upcoming sprint
- **In Progress** – Tasks currently being developed
- **Testing** – Tasks under verification and validation
- **Done** – Completed and approved tasks

Each user story is represented as a card and progresses across these columns based on its development stage. This structure ensures transparency and efficient task tracking.

39. Swimlanes Configuration

Swimlanes are configured to categorize tasks based on their nature, improving clarity and organization within the Agile board.

The following swimlanes are defined:

- **Feature Development** – Core functional requirements
- **Bug Fixes** – Issues identified during testing
- **Testing Activities** – Test case execution and validation
- **Documentation** – SRS, design, and reporting tasks

This classification enables the team to prioritize work effectively and maintain a clear distinction between development and supporting activities.

40. User Story Creation

User stories are developed following the **INVEST principle** (Independent, Negotiable, Valuable, Estimable, Small, Testable). These stories represent system requirements from the perspective of different stakeholders.

Examples of user stories include:

- As an HR recruiter, I want to create job openings so that hiring can begin efficiently.
- As an HR recruiter, I want to manage candidate information so that I can track applicants effectively.
- As an HR recruiter, I want to update candidate status so that recruitment progress is visible.
- As a hiring manager, I want to review candidate profiles so that I can make informed decisions.
- As a hiring manager, I want to provide interview feedback so that candidate evaluation is documented.
- As an HR recruiter, I want to schedule interviews so that candidates can be assessed systematically.

- As an admin, I want to manage user roles so that system access is controlled.
- As management, I want to generate reports so that recruitment performance can be analyzed.

A total of **15–20 user stories** are created to cover all major functionalities.

41. Epic Creation

User stories are grouped into high-level functional categories known as epics. This helps in organizing the backlog and aligning development with major system features.

The identified epics are:

- **Job Management**
- **Candidate Management**
- **Interview Management**
- **User Management**
- **Reporting and Analytics**

Each epic represents a major functional area of the Recruitment Management System.

42. Epic Breakdown

Each epic is further decomposed into smaller, manageable user stories.

For example:

Candidate Management Epic:

- Add candidate details
- Update candidate information
- Upload resumes
- Update recruitment status
- Search and filter candidates

Interview Management Epic:

- Schedule interviews
- Reschedule interviews
- Record interview feedback

This breakdown ensures that large functionalities are divided into implementable units.

43. Story Mapping Workshop

A story mapping workshop is conducted to visualize the system workflow and prioritize features based on user interaction.

The process includes:

1. Identifying the primary user (HR Recruiter)
2. Defining key activities such as managing jobs, candidates, interviews, and reports
3. Breaking activities into detailed user stories
4. Arranging stories in a logical sequence based on workflow

5. Prioritizing features for implementation

This approach ensures that development aligns with real-world usage scenarios.

44. Story Map Structure (MVP – Release 1)

The system is developed incrementally, starting with a **Minimum Viable Product (MVP)**.

Release 1 (MVP) includes:

- Job creation and management
- Candidate data entry and management
- Recruitment stage tracking
- Interview scheduling
- Basic reporting

Future releases may include:

- Advanced analytics
- Candidate ranking
- Notification systems

This phased approach ensures timely delivery of core functionalities.

45. Definition of Ready (DoR)

A user story is considered ready for development when it meets the following criteria:

- Clearly defined description
- Well-defined acceptance criteria
- No ambiguity in requirements
- Dependencies identified
- Estimated by the development team

This ensures that only well-understood tasks are taken into development.

46. Definition of Done (DoD)

A user story is considered complete when:

- Functionality is fully implemented
- Code is tested and verified
- No critical defects remain
- Code has been reviewed
- Documentation is updated

This ensures quality and consistency in deliverables.

47. Backlog Prioritization

The product backlog is prioritized based on business value and system importance.

High Priority:

- Job management
- Candidate management
- Recruitment tracking

Medium Priority:

- Interview scheduling
- Feedback management

Low Priority:

- Advanced reporting
- UI enhancements

Prioritization ensures that critical features are developed first, enabling early system usability.

Exercise 4: Sprint Planning & Backlog Grooming

48. Sprint Planning Meeting

A sprint planning meeting was conducted for a duration of 60 minutes to define the scope of the upcoming sprint for the Recruitment Management System. The meeting involved all team members, including the project manager, developers, and testers.

During the meeting, the product backlog was reviewed, and high-priority user stories were selected based on business value and feasibility. Each selected story was discussed in detail to ensure a clear understanding of requirements, dependencies, and expected outcomes.

The team also identified potential risks and clarified acceptance criteria before committing to the sprint.

49. Sprint Goal Definition

The sprint goal defines the primary objective to be achieved within the sprint duration.

Sprint Goal:

> To implement core functionalities of the Recruitment Management System, including job creation, candidate management, and recruitment stage tracking.

This goal ensures that the team focuses on delivering a functional and usable portion of the system.

50. Capacity Planning

Capacity planning was performed to estimate the total available working hours of the team during the sprint.

Team Member	Role	Available Hours/Day	Working Days	Total Capacity (Hours)
_____	Developer	6	10	_____
_____	Developer	6	10	_____
_____	Tester	5	10	_____
_____	Manager	4	10	_____

The total team capacity is calculated by summing individual capacities. This helps in determining how many user stories can be realistically completed within the sprint.

51. Story Point Estimation

Story point estimation was carried out using the **Planning Poker technique**, where each team member independently estimated the complexity of user stories.

The Fibonacci sequence (1, 2, 3, 5, 8) was used for estimation.

Example:

User Story	Description	Story Points
US-01	Create job opening	3
US-02	Add candidate details	5
US-03	Update candidate status	3
US-04	Schedule interview	5
US-05	Generate report	8

Consensus was reached through discussion, ensuring accurate and agreed-upon estimates.

52. Task Breakdown

Each user story was further divided into technical tasks with estimated effort in hours.

Example:

User Story: Add Candidate

Task	Description	Estimated Hours
Task 1	Design database schema	3
Task 2	Develop backend logic	5
Task 3	Create frontend form	4
Task 4	Integration testing	2

This breakdown improves task allocation and progress tracking.

53. Backlog Grooming

Backlog grooming was conducted to refine upcoming user stories and ensure they are ready for future sprints.

Activities performed:

- Reviewing incomplete or unclear stories
- Updating descriptions and acceptance criteria
- Removing redundant or low-priority tasks
- Re-prioritizing backlog items based on importance

This process ensures that the backlog remains organized and relevant.

54. Story Splitting

Large user stories were divided into smaller, manageable stories to improve development efficiency.

Example:

Original Story:
Manage Candidate Information

Split Into:

- Add candidate details
- Update candidate details
- Delete candidate records
- Search and filter candidates

Story splitting ensures that tasks are small, testable, and deliverable within a sprint.

55. Estimate Updates

During backlog grooming, story point estimates were reviewed and updated based on improved understanding of requirements.

Changes were made in the following cases:

- When new complexities were identified
- When scope of a story changed
- When dependencies were discovered

This ensures that sprint planning remains accurate and realistic.